

Modeling of a Permanent Magnet Synchronous Machine With Internal Magnets Using Magnetic Equivalent Circuits

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The design of control strategies for permanent magnet synchronous machines (PSM) is almost exclusively based on classical dq0-models. These models are, however, not able to systematically describe saturation or nonhomogenous air gap geometries typically occurring in PSM. This paper deals with a framework for the mathematical modeling of PSM based on magnetic equivalent circuits. Different to existing works, the model equations are derived by means of graph theory allowing for a systematic choice of a minimal set of state variables of the model and a systematic consideration of the electrical connection of the coils of the motor. The resulting model is calibrated and verified by means of measurement results. Finally, a magnetically linear and a dq0-model are derived and their performance is compared with the nonlinear model and measurement results.

Index Terms—Electric motor, magnetic equivalent circuit, permanent magnet motors.

I. INTRODUCTION

PERMANENT magnet synchronous motors (PSM) are widely used in many technical applications. Numerous papers and books dealing with the design of PSM have been published in recent years [1]–[7]. The mathematical models proposed in these papers range from finite element (FE) analysis over reluctance models to classical dq0-models. FE models exhibit a high accuracy for the calculated magnetic fields and allow for an exact consideration even of complex geometries of the motor [5]–[10]. Due to their high (numeric) complexity, these models are, however, hardly suitable for dynamical simulations and a controller design.

The design of control strategies for PSM is typically based on classical dq0-models, which, in their original form, assume a homogenous air gap and unsaturated iron cores [11]–[18]. Many modern designs of PSM (including, e.g., PSM with internal magnets) exhibit considerable saturation and nonsinusoidal fluxes in the coils. To cope with these effects, extensions of dq0-models have been reported in literature, which are all based on a heuristic approach and are limited to a very specific motor design [16]–[18]. In most cases, these models are not able to accurately describe the motor behavior in all operating conditions.

Magnetic equivalent circuits have become very popular for the design and the (dynamical) simulation of PSM in the recent years [1], [2], [19]–[35]. This is due to the significantly reduced complexity in comparison to FE models and their capability to systematically describe saturation and nonhomogenous air gap geometries. The accuracy and complexity of reluctance models can be easily controlled by means of the choice of the reluctance network. While reluctance networks with a rather high complexity are necessary to accurately describe field profiles in the motor, models of significantly reduced complexity already represent the behavior with respect to the torque, currents and voltages of the motor in sufficient

detail. Thus, models based on adequately chosen reluctance models promise to be a good basis for dynamical simulations and the (nonlinear) controller design.

In this paper, a framework for the systematic derivation of a state-space model with a minimum number of nonlinear equations and state variables is presented. The main purpose of the derived model is to provide a state-space representation for advanced model-based control strategies, and thus to reproduce the dynamic input-to-output behavior of the motor in an accurate manner. The framework developed here is universal, it is applied here to a specific internal magnet PSM that exhibits both large cogging torque and saturation. Section II presents the considered model and a complete reluctance model of the motor. To obtain a minimal set of (nonlinear) equations that describe the reluctance network, a method based on graph theory is proposed. This method, well known from electrical networks [36]–[38] is adjusted to the analysis of magnetic networks. Subsequently, the description of the electrical connection of the coils and the choice of a suitable set of state variables for the dynamical system is outlined. It should be noted that the framework presented in this section can be applied to any PSM. Section III is concerned with a reduced model based on findings of simulation results of the complete model. Section IV shows the systematic calibration of the reduced model and a comparison with measurement results. Starting from the nonlinear model, a magnetically linear model, and a classical dq0-model are systematically derived in Section V. Finally, the results of the nonlinear model, the magnetically linear model and the dq0-model are compared with measurement results.

II. CONSIDERED MOTOR AND COMPLETE MODEL

The motor considered in this paper is a permanent magnet synchronous motor with internal magnets. It comprises 12 stator coils, each wound around a single stator tooth, and eight NdFeB-magnets in the rotor, which are alternately magnetized. The setup of the motor is periodically repeated every 90° (number of pole pairs $p = 4$), such that only a quarter of the motor has to be considered. Fig. 1 shows a sectional view of the PSM and the permeance network used

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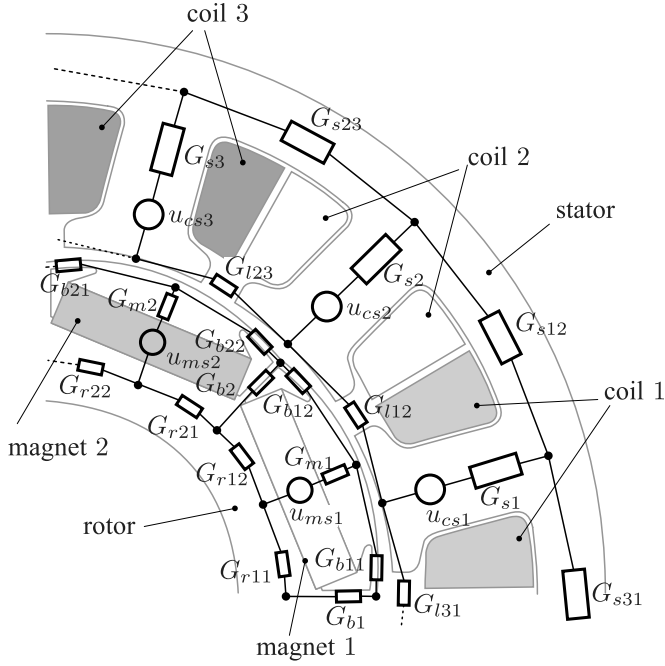


Fig. 1. Sectional view of the PSM with permeance network.

to model the stator and the rotor (air gap permeances are not included in this figure).

The motor is designed to exhibit a large cogging torque by means of an inhomogeneous construction of the air gap, Fig. 1. This is due to the fact that the motor is used in an application where external torques beyond a certain limit should not yield large changes in the rotor angle φ . The large cogging torque, however, makes the design of high-performance control strategies more involved. Thus, tailored mathematical models which accurately describe the cogging torque and the saturation are required for the controller design.

A. Permeance Network

As already outlined in Section I, a network of nonlinear permeances is utilized for the derivation of a model of the motor. Fig. 2 shows the proposed permeance network of the motor. The permeances describing the core of the stator and the rotor are approximated by cuboids of length l and area A . To account for saturation effects in the core, the relative permeability μ_r is defined as a function of the absolute value of the magnetic field strength $H = u/l$, i.e., $\mu_r(|u|/l)$, where u denotes the magnetomotive force. Fig. 3 shows the relative permeability μ_r for the applied core material M800-50A.

The nonlinear permeances of the stator teeth then read as

$$G_{sj}(u_{sj}) = \frac{A_{st}\mu_0\mu_r\left(\frac{|u_{sj}|}{l_{st}}\right)}{l_{st}} \quad j = 1, 2, 3 \quad (1)$$

with the area A_{st} , the length l_{st} and the magnetomotive force u_{sj} of a stator tooth, and the permeability μ_0 of free space. The permeances of the stator yoke can be found analogously

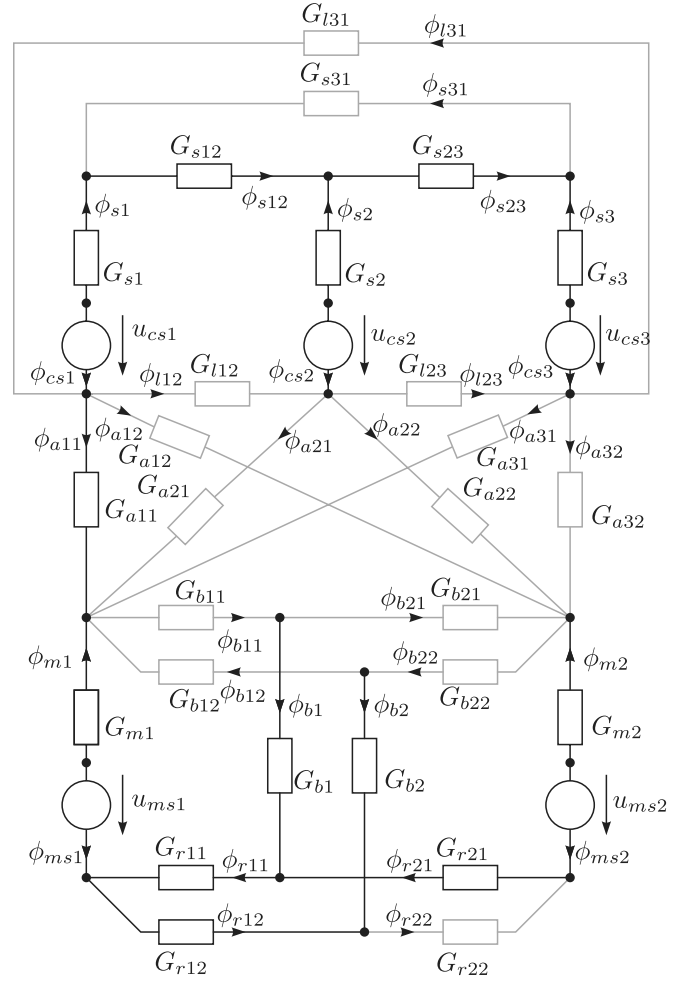


Fig. 2. Permeance network of the PSM.

in the form

$$G_{sjk}(u_{sjk}) = \frac{A_{sy}\mu_0\mu_r\left(\frac{|u_{sjk}|}{l_{sy}}\right)}{l_{sy}} \quad jk \in \{12, 23, 31\} \quad (2)$$

where A_{sy} is the area, l_{sy} describes the length, and u_{sjk} is the corresponding magnetomotive force. The center of the rotor is divided into four elements, which are described by the permeances

$$G_{rjk}(u_{rjk}) = \frac{A_r\mu_0\mu_r\left(\frac{|u_{rjk}|}{l_r}\right)}{l_r} \quad jk \in \{11, 12, 21, 22\}. \quad (3)$$

Here, A_r is the effective area, l_r the effective length, and u_{rjk} the magnetomotive force of the rotor element. The permanent magnets are placed inside the rotor of the motor. The resulting construction of the rotor exhibits parts, which have the form of very slender bars. The circumferential bars are described by

$$G_{bjk}(u_{bjk}) = \frac{A_{bc}\mu_0\mu_r\left(\frac{|u_{bjk}|}{l_{bc}}\right)}{l_{bc}} \quad jk \in \{11, 12, 21, 22\} \quad (4)$$

with the area A_{bc} , the length l_{bc} , and the magnetomotive force u_{bjk} . The permeances of the radial bars read as

$$G_{bj}(u_{bj}) = \frac{A_{br}\mu_0\mu_r\left(\frac{|u_{bj}|}{l_{br}}\right)}{l_{br}} \quad j = 1, 2. \quad (5)$$

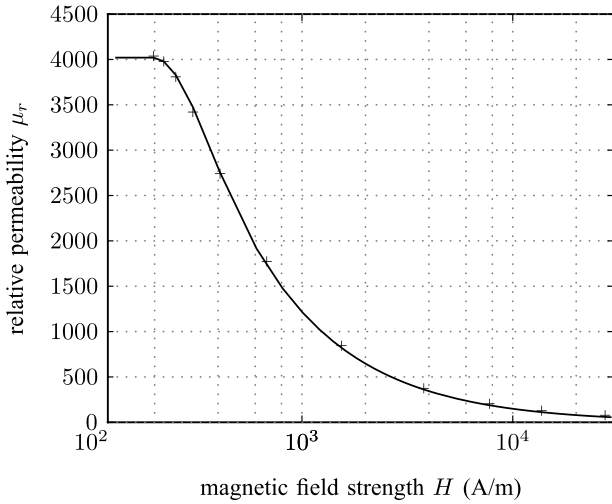


Fig. 3. Relative permeability μ_r of the core material M800-50A.

Again, A_{br} denotes the area, l_{br} the length, and u_{brj} the magnetomotive force of the radial bar element.

The air gap of the motor is modeled by two types of permeances: the permeances G_{ljk} , $jk \in \{12, 23, 31\}$, describing the leakage between adjacent stator teeth, and G_{ajk} , $jk \in \{11, 12, 21, 22, 31, 32\}$, describing the coupling between stator and rotor. The leakage permeances are defined as

$$G_{ljk} = \frac{A_l \mu_0}{l_l} \quad jk \in \{12, 23, 31\} \quad (6)$$

with the effective area A_l and length l_l . The air gap permeances G_{ajk} are, of course, functions of the relative rotation φ of the rotor with respect to the stator. A geometric model of these permeances using an approximate air gap geometry is possible but yields inaccurate results due to stray fluxes not covered by the approximate air gap geometry. Therefore, a heuristic approach, as has been proposed in [2] and [19], is used to approximate the coupling between the rotor and stator, i.e., the air gap permeance G_a

$$G_a(\varphi) = \begin{cases} 0 & -\frac{\pi}{4} \leq \tilde{\varphi} \leq -\delta \\ \frac{G_{a,\max}}{2} (1 + \cos(\frac{\pi}{\delta} \tilde{\varphi})) & -\delta < \tilde{\varphi} \leq \delta \\ 0 & \delta < \tilde{\varphi} \leq \frac{\pi}{4} \end{cases} \quad (7)$$

Therein, $\tilde{\varphi}$ is the relative rotation φ mapped to the interval $(-\pi/4, \pi/4)$ by means of a modulo operation. In addition, δ is a parameter which can be approximately determined by the geometrical overlap between a permanent magnet and a stator tooth, and $G_{a,\max}$ is the maximum value at $\tilde{\varphi} = 0$. Given G_a of (7), the air gap permeances between the individual stator teeth and permanent magnets are defined as

$$G_{ajk} = G_a \left(\varphi - \frac{(j-1)\pi}{6} - \frac{(k-1)\pi}{4} \right) \quad (8)$$

with $j = 1, 2, 3$ and $k = 1, 2$.

The NdFeB-magnets exhibit an almost linear behavior in the operating range, which can be modeled in the form of a constant magnetomotive force u_{msj} , $j = 1, 2$ and a linear

permeance

$$G_{mj}(u_{mj}) = \frac{A_m \mu_0 \mu_{rm}}{l_m} \quad j = 1, 2 \quad (9)$$

with the constant relative permeability μ_{rm} , the effective area A_m , and the length l_m . Given the coercive field strength H_c of the magnets, their magnetomotive forces are described by

$$u_{ms1} = -u_{ms2} = -H_c l_m. \quad (10)$$

The stator coils with N_c turns are modeled by

$$u_{csj} = N_c i_{cj} \quad j = 1, 2, 3 \quad (11)$$

with i_{cj} being the electric current through the coil j .

B. Balance Equations

Two approaches for the derivation of the balance equations (Kirchhoff's node and branch equations) are typically used for magnetic reluctance networks: 1) mesh analysis [33]–[35] and 2) node potential analysis [2], [19]–[21], [23], [25]–[28]. While a proper choice of meshes, yielding a set of independent equations might be tricky, the node potential analysis automatically guarantees the independence of the resulting equations. Therefore, node potential analysis is typically favored.

In this paper, an alternative approach for the systematic derivation of a minimal set of independent equations based on graph theory is proposed. It uses a tree, which connects all nodes of the network without forming any meshes. This approach is well known from electric network analysis [36]–[38], and can be, as will be shown in this paper, directly applied to magnetic permeance networks, see also [29].

The chosen tree has to connect all nodes of the network without forming any meshes. In addition, all magnetomotive force sources have to be included in the tree, which is always possible for nondegenerated networks. It further turns out to be advantageous to exclude as many air gap permeances from the tree as possible. One possible choice of a tree is given in Fig. 2 by the components depicted in black. The cotree is then composed of all components which are not part of the tree (depicted gray in Fig. 2). Adding one cotree element to the tree yields a single mesh.

For the subsequent derivation, it is useful to subdivide the elements of the tree into magnetomotive force sources of the coils (index tc), magnetomotive force sources of the permanent magnets (index tm), and permeances (linear, nonlinear, angle dependent, index tg). Then, the overall vector of the tree fluxes $\phi_t = [\phi_{tc}^T, \phi_{tm}^T, \phi_{tg}^T]^T$ is defined by

$$\phi_{tc} = [\phi_{cs1}, \phi_{cs2}, \phi_{cs3}]^T \quad (12a)$$

$$\phi_{tm} = [\phi_{ms1}, \phi_{ms2}]^T \quad (12b)$$

$$\phi_{tg} = [\phi_{s1}, \phi_{s2}, \phi_{s3}, \phi_{s12}, \phi_{s23}, \phi_{b1}, \phi_{b2}, \phi_{r11}, \phi_{r12}, \phi_{r21}, \phi_{m1}, \phi_{m2}, \phi_{a11}]^T. \quad (12c)$$

The vector of the corresponding tree magnetomotive forces $\mathbf{u}_t$ is defined in an analogous manner.

The cotree only comprises permeances such that the vector of the cotree fluxes is given by

$$\boldsymbol{\phi}_c = [\phi_{l12}, \phi_{l23}, \phi_{l31}, \phi_{s31}, \phi_{b11}, \phi_{b12}, \phi_{b21}, \phi_{b22}, \phi_{r22}, \phi_{a12}, \phi_{a21}, \phi_{a22}, \phi_{a31}, \phi_{a32}]^T \quad (13)$$

and the vector of cotree magnetomotive forces $\mathbf{u}_c$ is defined in the same way. Now, the following relations between the tree and cotree fluxes and magnetomotive forces, respectively, can be formulated:

$$\boldsymbol{\phi}_t = \mathbf{D}\boldsymbol{\phi}_c \quad (14a)$$

$$\mathbf{u}_c = -\mathbf{D}^T \mathbf{u}_t. \quad (14b)$$

The incidence matrix $\mathbf{D}$ describes the interconnection of the individual elements of the permeance network and its entries are either -1 , 0 , or 1 . It can be decomposed into a part $\mathbf{D}_c$ linking the cotree fluxes with the tree coil fluxes, a part $\mathbf{D}_m$ linking the cotree fluxes with the tree permanent magnet fluxes, and a part $\mathbf{D}_g$, which connects the cotree fluxes with the fluxes of the tree permeances, i.e., $\mathbf{D}^T = [\mathbf{D}_c^T, \mathbf{D}_m^T, \mathbf{D}_g^T]$.

The constitutive equations of the permeances can be summarized in the form

$$\boldsymbol{\phi}_{tg} = \mathbf{G}_t \mathbf{u}_{tg} \quad (15a)$$

$$\boldsymbol{\phi}_c = \mathbf{G}_c \mathbf{u}_c \quad (15b)$$

with the permeance matrices $\mathbf{G}_t$ and $\mathbf{G}_c$ of the tree and cotree, respectively. Note that in general these matrices are functions of the corresponding magnetomotive forces (due to saturation) and the displacement of the rotor, i.e., $\mathbf{G}_t(\mathbf{u}_{tg}, \varphi)$ and $\mathbf{G}_c(\mathbf{u}_c, \varphi)$. For the permeance network of Fig. 2 these matrices read as

$$\mathbf{G}_t = \text{diag}[G_{s1}, G_{s2}, G_{s3}, G_{s12}, G_{s23}, G_{b1}, G_{b2}, G_{r11}, G_{r12}, G_{r21}, G_{m1}, G_{m2}, G_{a11}] \quad (16a)$$

$$\mathbf{G}_c = \text{diag}[G_{l12}, G_{l23}, G_{l31}, G_{s31}, G_{b11}, G_{b12}, G_{b21}, G_{b22}, G_{r22}, G_{a12}, G_{a21}, G_{a22}, G_{a31}, G_{a32}]. \quad (16b)$$

Inserting (15) into (14), we find the following set of equations:

$$\begin{bmatrix} \boldsymbol{\phi}_{tc} \\ \boldsymbol{\phi}_{tm} \\ \mathbf{G}_t \mathbf{u}_{tg} \end{bmatrix} = -\mathbf{D}\mathbf{G}_c [\mathbf{D}_c^T, \mathbf{D}_m^T, \mathbf{D}_g^T] \begin{bmatrix} \mathbf{u}_{tc} \\ \mathbf{u}_{tm} \\ \mathbf{u}_{tg} \end{bmatrix}. \quad (17)$$

If it is assumed that the coil currents $\mathbf{i}_c = [i_{c1}, i_{c2}, i_{c3}]^T$ and thus the magnetomotive forces $\mathbf{u}_{tc}$ are given, the unknown variables of (17) are $\boldsymbol{\phi}_{tc}$, $\boldsymbol{\phi}_{tm}$ and $\mathbf{u}_{tg}$. A simple reformulation of (17) yields

$$\begin{bmatrix} \mathbf{I} & \mathbf{0} & \mathbf{D}_c \mathbf{G}_c \mathbf{D}_c^T \\ \mathbf{0} & \mathbf{I} & \mathbf{D}_m \mathbf{G}_c \mathbf{D}_m^T \\ \mathbf{0} & \mathbf{0} & \mathbf{G}_t + \mathbf{D}_g \mathbf{G}_c \mathbf{D}_g^T \end{bmatrix} \begin{bmatrix} \boldsymbol{\phi}_{tc} \\ \boldsymbol{\phi}_{tm} \\ \mathbf{u}_{tg} \end{bmatrix} = -\mathbf{D}\mathbf{G}_c (\mathbf{D}_c^T \mathbf{u}_{tc} + \mathbf{D}_m^T \mathbf{u}_{tm}) \quad (18)$$

with the identity matrix $\mathbf{I}$. It can be easily seen that a set of $\dim(\mathbf{u}_{tg}) = n = 13$ nonlinear algebraic equations has to be solved for $\mathbf{u}_{tg}$. All other quantities of the network can be calculated from simple linear equations. A proof of the existence and uniqueness of a solution of the nonlinear algebraic equations (18) is given in the Appendix.

C. Torque Equation

Starting from the magnetic coenergy of the permeance network, the electromagnetic torque of the motor is defined as

$$\tau = \frac{1}{2} p \left(\mathbf{u}_{tg}^T \frac{\partial \mathbf{G}_t}{\partial \varphi} \mathbf{u}_{tg} + \mathbf{u}_c^T \frac{\partial \mathbf{G}_c}{\partial \varphi} \mathbf{u}_c \right) \quad (19)$$

with the number p of pole pairs [2]. With the help of (14b) this equation can be reformulated in the form

$$\tau = \frac{1}{2} p \left(\mathbf{u}_{tg}^T \frac{\partial \mathbf{G}_t}{\partial \varphi} \mathbf{u}_{tg} + \mathbf{u}_t^T \mathbf{D} \frac{\partial \mathbf{G}_c}{\partial \varphi} \mathbf{D}^T \mathbf{u}_t \right) \quad (20)$$

$$\text{with } \mathbf{u}_t^T = [\mathbf{u}_{tc}^T, \mathbf{u}_{tm}^T, \mathbf{u}_{tg}^T].$$

D. Voltage Equation

The mathematical model (18) and (20) allows for a calculation of the magnetomotive forces, fluxes and the torque for given currents $\mathbf{i}_c$. This model is useful for a static analysis of the motor. In a dynamical analysis, however, the coil voltages $\mathbf{v}_c$ must be used as inputs. This relation is provided by Faraday's law

$$\frac{d\boldsymbol{\psi}_c}{dt} = \mathbf{R}_c \mathbf{i}_c - \mathbf{v}_c \quad (21)$$

with the flux linkage $\boldsymbol{\psi}_c = \mathbf{N}_c \boldsymbol{\phi}_{tc}$, the winding matrix $\mathbf{N}_c = \text{diag}[N_c, N_c, N_c]$, the electric resistance matrix $\mathbf{R}_c = \text{diag}[R_c, R_c, R_c]$, and the electric voltages $\mathbf{v}_c = [v_{c1}, v_{c2}, v_{c3}]^T$. Here, N_c is the number of turns, R_c the electric resistance, and v_{cj} the voltage of the respective coil $j = 1, 2, 3$. Equation (21) links the fluxes $\boldsymbol{\phi}_{tc}$ of the coils with their currents $\mathbf{i}_c$. Thus, either $\boldsymbol{\phi}_{tc}$ has to be defined as a function of $\mathbf{i}_c$ or vice versa. For nonlinear permeance networks, it proves to be advantageous to express the coil currents $\mathbf{i}_c$ as functions of the fluxes by reformulating (18) in the form

$$\underbrace{\begin{bmatrix} \mathbf{D}_c \mathbf{G}_c \mathbf{D}_c^T & \mathbf{0} & \mathbf{D}_c \mathbf{G}_c \mathbf{D}_g^T \\ \mathbf{D}_m \mathbf{G}_c \mathbf{D}_c^T & \mathbf{I} & \mathbf{D}_m \mathbf{G}_c \mathbf{D}_g^T \\ \mathbf{D}_g \mathbf{G}_c \mathbf{D}_c^T & \mathbf{0} & \mathbf{G}_t + \mathbf{D}_g \mathbf{G}_c \mathbf{D}_g^T \end{bmatrix}}_{\mathbf{K}_1} \underbrace{\begin{bmatrix} \mathbf{u}_{tc} \\ \boldsymbol{\phi}_{tm} \\ \mathbf{u}_{tg} \end{bmatrix}}_{\mathbf{x}} = - \underbrace{\begin{bmatrix} \boldsymbol{\phi}_{tc} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}}_{\mathbf{M}_1} - \underbrace{\mathbf{D}\mathbf{G}_c \mathbf{D}_m^T \mathbf{u}_{tm}}_{\mathbf{M}_2}. \quad (22)$$

This means that the dynamical model of the motor is given by a set of nonlinear differential-algebraic equations (DAE), i.e., (21) and (22). Now, the following questions arise.

- 1) Do the state variables of (21) represent the minimum number of states or is it possible to reduce the number of states?
- 2) Does the nonlinear set of equations (22) have a unique solution?
- 3) How can the electric interconnection of the coils (e.g., delta or wye connection) be systematically considered?

To answer the first two questions consider the matrix $\mathbf{K}_1$ of (22). Using the results of the Appendix, it turns out that $\mathbf{K}_1$ is singular if the rows of $\mathbf{D}_c$ are linearly dependent.

Let us assume that $\mathbf{D}_c \in \mathbb{R}^{m \times n}$, $m < n$ has $m^\perp$ linear dependent rows. Then, the column space $\mathcal{D}_c^I = \text{span}(\mathbf{D}_c)$ has dimension $m - m^\perp$ and the orthogonal complement $\mathcal{D}_c^\perp = \text{span}(\mathbf{a} \in \mathbb{R}^m | \mathbf{a}^T \mathbf{b} = 0, \forall \mathbf{b} \in \mathcal{D}_c^I)$ has dimension $m^\perp$. Let $\mathbf{D}_c^I$ be a matrix composed of $m - m^\perp$ independent vectors of $\mathcal{D}_c^I$ (i.e., the image of $\mathbf{D}_c$) and $\mathbf{D}_c^\perp$ be composed of $m^\perp$ independent vectors of $\mathcal{D}_c^\perp$ (i.e., the kernel of $\mathbf{D}_c^T$). Then, $(\mathbf{D}_c^\perp)^T \mathbf{D}_c = \mathbf{0}$ holds and the nonsingular matrix

$$\mathbf{T}_1 = \begin{bmatrix} \mathbf{T}_{1c} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix} \quad (23)$$

with

$$\mathbf{T}_{1c} = \begin{bmatrix} (\mathbf{D}_c^\perp)^T \\ (\mathbf{D}_c^I)^T \end{bmatrix} \quad (24)$$

can be defined. Applying the transformation matrix $\mathbf{T}_1$ in the form

$$\underbrace{\mathbf{T}_1 \mathbf{K}_1 \mathbf{T}_1^{-1}}_{\mathbf{K}_2} \mathbf{T}_1 \begin{bmatrix} \mathbf{u}_{tc} \\ \boldsymbol{\phi}_{tm} \\ \mathbf{u}_{tg} \end{bmatrix} = -\mathbf{T}_1 \mathbf{M}_1 - \mathbf{T}_1 \mathbf{M}_2 \quad (25)$$

results in a matrix $\mathbf{K}_2$ with the structure

$$\mathbf{K}_2 = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_{2r} \end{bmatrix} \quad (26)$$

where the number of zero rows and columns is $m^\perp$. To prove this statement, $\mathbf{K}_2$ is formulated as

$$\mathbf{K}_2 = \begin{bmatrix} \mathbf{T}_{1c} \mathbf{D}_c \mathbf{G}_c \mathbf{D}_c^T \mathbf{T}_{1c}^{-1} & \mathbf{0} & \mathbf{T}_{1c} \mathbf{D}_c \mathbf{G}_c \mathbf{D}_g^T \\ \mathbf{D}_m \mathbf{G}_c \mathbf{D}_c^T \mathbf{T}_{1c}^{-1} & \mathbf{I} & \mathbf{D}_m \mathbf{G}_c \mathbf{D}_g^T \\ \mathbf{D}_g \mathbf{G}_c \mathbf{D}_c^T \mathbf{T}_{1c}^{-1} & \mathbf{0} & \mathbf{G}_t + \mathbf{D}_g \mathbf{G}_c \mathbf{D}_g^T \end{bmatrix}. \quad (27)$$

It can be seen that the product

$$\mathbf{T}_{1c} \mathbf{D}_c = \begin{bmatrix} \mathbf{0} \\ (\mathbf{D}_c^I)^T \mathbf{D}_c \end{bmatrix} \quad (28)$$

gives $m^\perp$ zero rows. Of course, the right-hand side multiplication with an arbitrary matrix does not change the zero rows. To prove the zero columns in $\mathbf{K}_2$, the product $\mathbf{D}_c^T \mathbf{T}_{1c}^{-1}$ is analyzed. The matrix $\mathbf{T}_{1c}$ can be written in the form

$$\mathbf{T}_{1c} = \begin{bmatrix} \mathbf{a}_1^T \\ \vdots \\ \mathbf{a}_{m^\perp}^T \\ \mathbf{b}_1^T \\ \vdots \\ \mathbf{b}_{m-m^\perp}^T \end{bmatrix} \quad (29)$$

where $\mathbf{a}_j \in \mathcal{D}_c^\perp$ and $\mathbf{b}_j \in \mathcal{D}_c^I$. The inverse $\mathbf{T}_{1c}^{-1} = [\mathbf{v}_1, \dots, \mathbf{v}_{m^\perp}, \mathbf{w}_1, \dots, \mathbf{w}_{m-m^\perp}]$ has to meet $\mathbf{T}_{1c} \mathbf{T}_{1c}^{-1} = \mathbf{I}$ and therefore

$$\mathbf{a}_j^T \mathbf{v}_k = \delta_{jk} \quad \mathbf{a}_j^T \mathbf{w}_k = 0 \quad (30a)$$

$$\mathbf{b}_j^T \mathbf{w}_k = \delta_{jk} \quad \mathbf{b}_j^T \mathbf{v}_k = 0 \quad (30b)$$

with the Kronecker symbol δ_{jk} , holds. Obviously, this means that $\mathbf{v}_j \in \mathcal{D}_c^\perp$ and $\mathbf{w}_j \in \mathcal{D}_c^I$. Based on this discussion

$$\mathbf{D}_c^T \mathbf{T}_{1c}^{-1} = [\mathbf{0}, \star] \quad (31)$$

holds, where the number of zero columns is equal to $m^\perp$ and $\star$ is a matrix with $m - m^\perp$ nonzero columns. Thus, $\mathbf{K}_2$ has $m^\perp$ zero columns and rows.

The application of $\mathbf{T}_1$ to the vector of unknowns gives

$$\mathbf{T}_1 \begin{bmatrix} \mathbf{u}_{tc} \\ \boldsymbol{\phi}_{tm} \\ \mathbf{u}_{tg} \end{bmatrix} = \begin{bmatrix} \mathbf{T}_{1c} \mathbf{u}_{tc} \\ \boldsymbol{\phi}_{tm} \\ \mathbf{u}_{tg} \end{bmatrix} \quad (32)$$

the multiplication of $\mathbf{M}_1$ with the transformation matrix results in

$$\mathbf{T}_1 \begin{bmatrix} \boldsymbol{\phi}_{tc} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{T}_{1c} \boldsymbol{\phi}_{tc} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} \quad (33)$$

and $\mathbf{T}_1$ used in combination with $\mathbf{M}_2$ yields

$$\mathbf{T}_1 \mathbf{D} \mathbf{G}_c \mathbf{D}_g^T \mathbf{u}_{tm} = \begin{bmatrix} \mathbf{T}_{1c} \mathbf{D}_c \\ \mathbf{D}_m \\ \mathbf{D}_g \end{bmatrix} \mathbf{G}_c \mathbf{D}_g^T \mathbf{u}_{tm} \quad (34)$$

which again has $m^\perp$ zero rows.

This discussion shows two important results: 1) from (25) with (32) and (24) it can be seen that the part $(\mathbf{D}_c^\perp)^T \mathbf{u}_{tc}$ of the coil currents cannot be calculated from the permeance network but has to be defined by the electrical connection of the coils. Only the part $(\mathbf{D}_c^I)^T \mathbf{u}_{tc}$ is determined by the (reduced) set of nonlinear equations

$$\mathbf{K}_{2r} \begin{bmatrix} (\mathbf{D}_c^I)^T \mathbf{u}_{tc} \\ \boldsymbol{\phi}_{tm} \\ \mathbf{u}_{tg} \end{bmatrix} = - \begin{bmatrix} (\mathbf{D}_c^I)^T \boldsymbol{\phi}_{tc} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} - \begin{bmatrix} (\mathbf{D}_c^I)^T \mathbf{D}_c \\ \mathbf{D}_m \\ \mathbf{D}_g \end{bmatrix} \mathbf{G}_c \mathbf{D}_g^T \mathbf{u}_{tm}. \quad (35)$$

2) Not the entire part of $\boldsymbol{\phi}_{tc}$ is independent but the components are restricted to fulfill

$$(\mathbf{D}_c^\perp)^T \boldsymbol{\phi}_{tc} = \mathbf{0}. \quad (36)$$

This implies that the set of differential equations (21) for the flux linkage can be reduced to $m - m^\perp$ differential equations, where $(\mathbf{D}_c^I)^T \boldsymbol{\phi}_{tc}$ is a possible choice of independent states.

Remark 1: To systematically obtain the reduced set of nonlinear equations from the transformed set of equations (25), the reduction matrix $\mathbf{H}_1$

$$\mathbf{H}_1 = \begin{bmatrix} \mathbf{H}_{1r} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix} \quad (37)$$

$\mathbf{H}_1 \mathbf{H}_1^T = \mathbf{I}$ with $\mathbf{H}_{1r} = [\mathbf{0}, \mathbf{I}] \in \mathbb{R}^{(m-m^\perp) \times m}$, is introduced. Multiplying (25) with $\mathbf{H}_1$ from the left side directly yields the reduced equations

$$\underbrace{\mathbf{H}_1 \mathbf{K}_2 \mathbf{H}_1^T}_{\mathbf{K}_{2r}} \begin{bmatrix} \mathbf{H}_{1r} \mathbf{T}_{1c} \mathbf{u}_{tc} \\ \boldsymbol{\phi}_{tm} \\ \mathbf{u}_{tg} \end{bmatrix} = - \begin{bmatrix} \mathbf{H}_{1r} \mathbf{T}_{1c} \boldsymbol{\phi}_{tc} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} - \mathbf{H}_1 \mathbf{T}_1 \mathbf{M}_2 \quad (38)$$

with

$$\mathbf{K}_{2r} = \begin{bmatrix} \mathbf{H}_{1r}\mathbf{T}_{1c}\mathbf{D}_c\mathbf{G}_c\mathbf{D}_c^T\mathbf{T}_{1c}^{-1}\mathbf{H}_{1r}^T & \mathbf{0} & \mathbf{H}_{1r}\mathbf{T}_{1c}\mathbf{D}_c\mathbf{G}_c\mathbf{D}_g^T \\ \mathbf{D}_m\mathbf{G}_c\mathbf{D}_c^T\mathbf{T}_{1c}^{-1}\mathbf{H}_{1r}^T & \mathbf{I} & \mathbf{D}_m\mathbf{G}_c\mathbf{D}_g^T \\ \mathbf{D}_g\mathbf{G}_c\mathbf{D}_c^T\mathbf{T}_{1c}^{-1}\mathbf{H}_{1r}^T & \mathbf{0} & \mathbf{G}_t + \mathbf{D}_g\mathbf{G}_c\mathbf{D}_g^T \end{bmatrix} \quad (39)$$

and the new vector of unknowns

$$\mathbf{H}_1\mathbf{T}_1 \begin{bmatrix} \mathbf{u}_{tc} \\ \boldsymbol{\phi}_{tm} \\ \mathbf{u}_{tg} \end{bmatrix} = \begin{bmatrix} \mathbf{H}_{1r}\mathbf{T}_{1c}\mathbf{u}_{tc} \\ \boldsymbol{\phi}_{tm} \\ \mathbf{u}_{tg} \end{bmatrix}. \quad (40)$$

In a further step, the electrical connection of the coils will be considered by means of the interconnection matrix $\mathbf{V}_c$, that is

$$\mathbf{u}_{tc} = \mathbf{V}_c \bar{\mathbf{u}}_{tc}. \quad (41)$$

Here, $\bar{\mathbf{u}}_{tc}$ corresponds to the independent currents of the coils. Using, e.g., a wye connection of the three coils, the constraint reads as $i_{c1} + i_{c2} + i_{c3} = 0$, which can be accounted for by the interconnection matrix

$$\begin{bmatrix} i_{c1} \\ i_{c2} \\ i_{c3} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} i_{c1} \\ i_{c2} \end{bmatrix}. \quad (42)$$

Thus, $\bar{\mathbf{u}}_{tc} = N_c [i_{c1}, i_{c2}]^T$ has been chosen as the vector of independent currents. Replacing $\mathbf{u}_{tc}$ by (41) in the reduced vector of unknowns (40) results in

$$\begin{bmatrix} \mathbf{H}_{1r}\mathbf{T}_{1c}\mathbf{u}_{tc} \\ \boldsymbol{\phi}_{tm} \\ \mathbf{u}_{tg} \end{bmatrix} = \begin{bmatrix} \mathbf{H}_{1r}\mathbf{T}_{1c}\mathbf{V}_c\bar{\mathbf{u}}_{tc} \\ \boldsymbol{\phi}_{tm} \\ \mathbf{u}_{tg} \end{bmatrix}. \quad (43)$$

If the matrix $\mathbf{H}_{1r}\mathbf{T}_{1c}\mathbf{V}_c$ is nonsingular, $\bar{\mathbf{u}}_{tc}$ can be used as the new vector of independent unknown coil currents and no further action is necessary. In cases where the matrix is not square, the resulting nonlinear set of equations is overdetermined, i.e., there are more equations than unknowns. This can directly be seen by calculating the left-hand side of the reduced set of equations (38) in the form $\mathbf{K}_3 [\bar{\mathbf{u}}_{tc}^T, \boldsymbol{\phi}_{tm}^T, \mathbf{u}_{tg}^T]^T$, with $\mathbf{K}_3$ given by

$$\mathbf{K}_3 = \begin{bmatrix} \mathbf{S}_{11} & \mathbf{0} & \mathbf{H}_{1r}\mathbf{T}_{1c}\mathbf{D}_c\mathbf{G}_c\mathbf{D}_g^T \\ \mathbf{S}_{21} & \mathbf{I} & \mathbf{D}_m\mathbf{G}_c\mathbf{D}_g^T \\ \mathbf{S}_{31} & \mathbf{0} & \mathbf{G}_t + \mathbf{D}_g\mathbf{G}_c\mathbf{D}_g^T \end{bmatrix} \quad (44)$$

where

$$\mathbf{S}_{11} = \mathbf{H}_{1r}\mathbf{T}_{1c}\mathbf{D}_c\mathbf{G}_c\mathbf{D}_c^T\mathbf{T}_{1c}^{-1}\mathbf{H}_{1r}^T\mathbf{H}_{1r}\mathbf{T}_{1c}\mathbf{V}_c \quad (45a)$$

$$\mathbf{S}_{21} = \mathbf{D}_m\mathbf{G}_c\mathbf{D}_c^T\mathbf{T}_{1c}^{-1}\mathbf{H}_{1r}^T\mathbf{H}_{1r}\mathbf{T}_{1c}\mathbf{V}_c \quad (45b)$$

$$\mathbf{S}_{31} = \mathbf{D}_g\mathbf{G}_c\mathbf{D}_c^T\mathbf{T}_{1c}^{-1}\mathbf{H}_{1r}^T\mathbf{H}_{1r}\mathbf{T}_{1c}\mathbf{V}_c. \quad (45c)$$

Under the previous assumption that $\mathbf{H}_{1r}\mathbf{T}_{1c}\mathbf{V}_c$ is not square, the matrix $\mathbf{K}_3$ has more rows than columns, which implies that not all components of the reduced flux vector $\mathbf{H}_{1r}\mathbf{T}_{1c}\boldsymbol{\phi}_{tc}$ in (38) can be arbitrarily assigned and therefore used as state variables in (21). Thus, a part of the reduced flux vector has to be added to the vector of unknowns. Let us assume that the upper-left entry $\mathbf{S}_{11}$ of $\mathbf{K}_3$ has $n^\perp$ dependent rows. The transformation $\mathbf{T}_2 = [\mathbf{S}_{11}^\perp, \mathbf{S}_{11}^I]$, where $\mathbf{S}_{11}^I$ is the column space

of $\mathbf{S}_{11}$ and $\mathbf{S}_{11}^\perp$ is the orthogonal complement to $\mathbf{S}_{11}^I$, is used to introduce a transformed vector $\tilde{\boldsymbol{\phi}}_{tc}$ in the form

$$\mathbf{T}_2\tilde{\boldsymbol{\phi}}_{tc} = \begin{pmatrix} \mathbf{S}_{11}^\perp [\mathbf{I}, \mathbf{0}] + \mathbf{S}_{11}^I [\mathbf{0}, \mathbf{I}] \\ \mathbf{H}_{3r} \quad \mathbf{H}_{4r} \end{pmatrix} \tilde{\boldsymbol{\phi}}_{tc} = \mathbf{H}_{1r}\mathbf{T}_{1c}\boldsymbol{\phi}_{tc}. \quad (46)$$

It can be seen that adding the first $n^\perp$ elements of $\tilde{\boldsymbol{\phi}}_{tc}$ to the vector of unknowns results in a set of nonlinear equations with a unique solution. To do so, (46) is inserted into (38) with (41), (44), and (45) resulting in

$$[\mathbf{S}_2 \quad \mathbf{K}_3] \begin{bmatrix} \mathbf{H}_{3r}\tilde{\boldsymbol{\phi}}_{tc} \\ \bar{\mathbf{u}}_{tc} \\ \boldsymbol{\phi}_{tm} \\ \mathbf{u}_{tg} \end{bmatrix} = - \begin{bmatrix} \mathbf{S}_{11}^I \mathbf{H}_{4r}\tilde{\boldsymbol{\phi}}_{tc} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} - \mathbf{H}_1\mathbf{T}_1\mathbf{M}_2 \quad (47)$$

with $\mathbf{S}_2 = [(\mathbf{S}_{11}^\perp)^T, \mathbf{0}, \mathbf{0}]^T$. Obviously, $\mathbf{H}_{3r}\tilde{\boldsymbol{\phi}}_{tc}$ is obtained as a solution of (47) and $\mathbf{H}_{4r}\tilde{\boldsymbol{\phi}}_{tc}$ has to be used as independent state in the dynamical equation [see (21), (38), and (41)]

$$\frac{d}{dt}\mathbf{H}_{4r}\tilde{\boldsymbol{\phi}}_{tc} = \mathbf{H}_{4r}\mathbf{T}_2^{-1}\mathbf{H}_{1r}\mathbf{T}_{1c}\mathbf{N}_c^{-1}(\mathbf{R}_c\mathbf{N}_c^{-1}\mathbf{V}_c\bar{\mathbf{u}}_{tc} - \mathbf{v}_c). \quad (48)$$

As a result of this modeling framework we get the DAE system (47), (48) which is of minimal dimension and systematically accounts for the electric interconnection of the coils.

E. Simulation Results

To evaluate the behavior of the PSM, simulations of the mathematical model were performed. In a first step, the torque and the magnetomotive forces for fixed currents were investigated using (18) and (20). Fig. 4(a) shows the cogging torque, i.e., the torque for zero currents $i_{cj} = 0$, $j = 1, 2, 3$. It can be seen that a pronounced cogging torque with a periodicity of 15° is present in the motor. The results given in Fig. 2(b)–(d) were obtained for $-i_{c2} = i_{c3} = 2.5$ A, $i_{c1} = 0$ A, which approximately corresponds to the nominal value. A closer look at the torque in Fig. 4(b) shows that the characteristics of the torque is far from being sinusoidal, which would be expected for an ideal PSM. The magnetomotive forces in the stator teeth and yoke depicted in Fig. 4(c) and (d) further reveal that the magnetomotive forces in the yoke are much smaller than for the teeth.

This fact gives rise to the development of a simplified permeance network, which covers the essential effects of the complete model. A simplified model of reduced dimension and complexity is especially desirable for a prospective controller design. Thus, the following simplifications will be made: 1) the permeances G_{s12} , G_{s23} , and G_{s31} of the stator yoke are neglected, i.e., set to ∞ . 2) Simulations show that the fluxes in the radial rotor bars are very small compared to the fluxes in the rest of the motor. Thus, the simplification $G_{b1} = G_{b2} = 0$ is used. 3) With the last simplification, the circumferential rotor bars and the center of the rotor can be modeled by a single equivalent permeance G_b and G_r , respectively.

In the subsequent section, the simplified model will be presented in more detail. A comparison of simulation results of the complete with the reduced model will justify the simplifying assumptions being made.

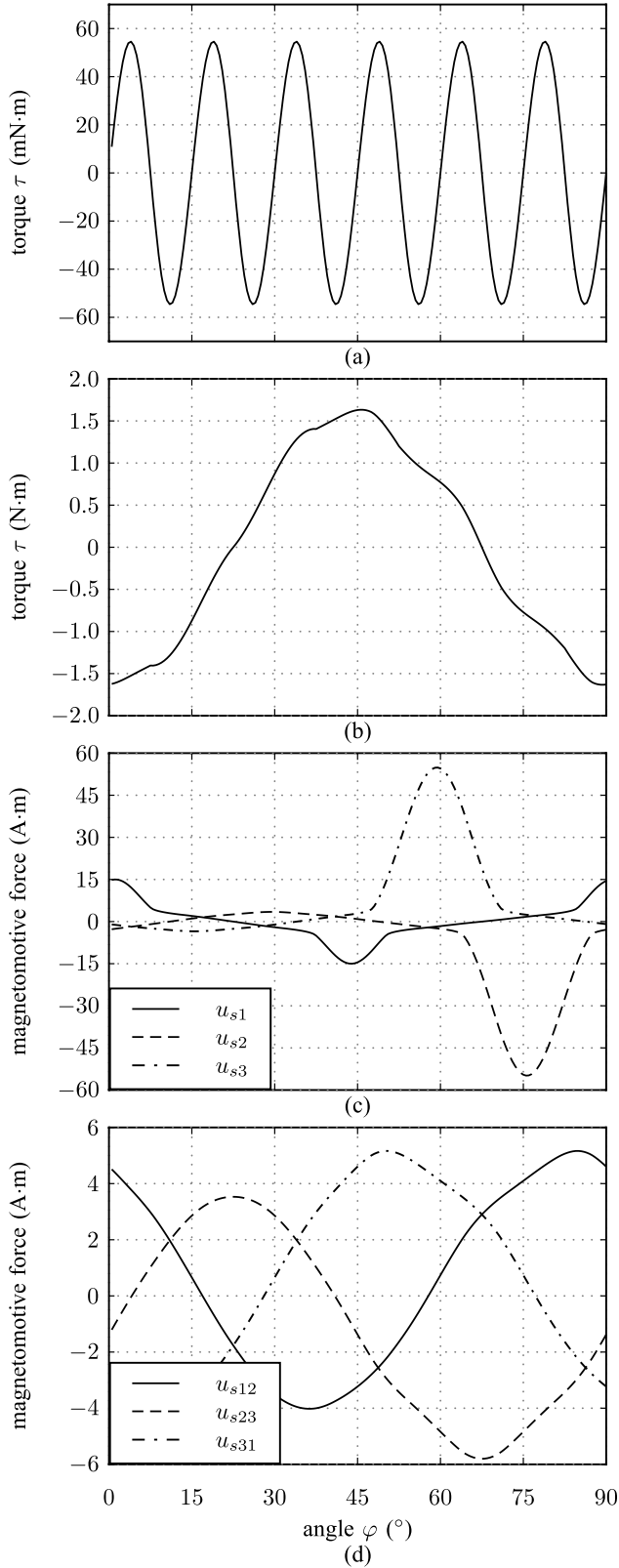


Fig. 4. Simulation results of the complete model (a) for zero currents and (b)–(d) for $-i_{c2} = i_{c3} = 2.5$ A, $i_{c1} = 0$ A.

III. REDUCED MODEL

A. Permeance Network

Fig. 5 shows the reduced permeance network. Therein, the effective permeances of the circumferential bars and the center

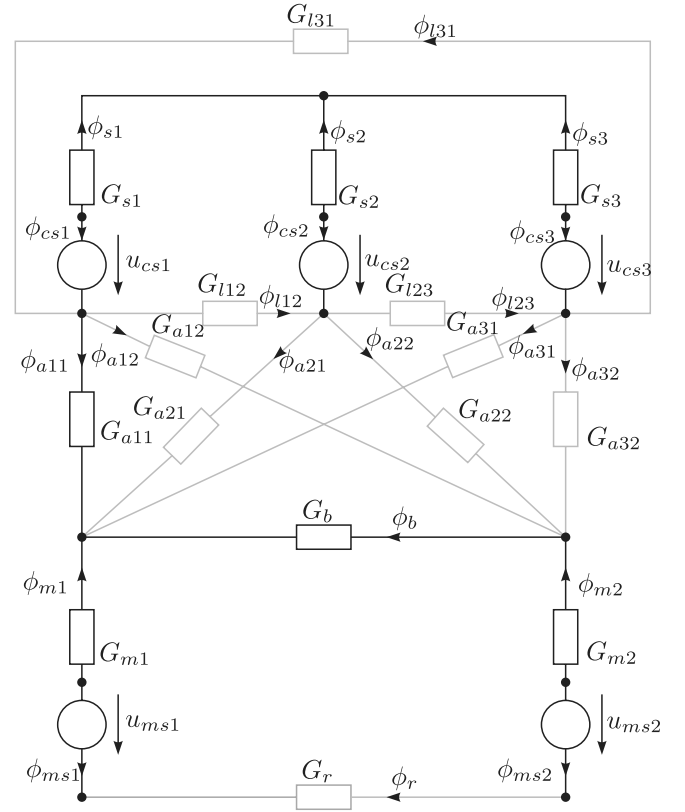


Fig. 5. Reduced permeance network of the PSM.

of the rotor are given by

$$G_b = \frac{A_{bc} \mu_0 \mu_r \left(\frac{|u_b|}{2l_{bc}} \right)}{l_{bc}} \quad (49a)$$

$$G_r = \frac{A_r \mu_0 \mu_r \left(\frac{|u_r|}{2l_r} \right)}{l_r} \quad (49b)$$

while all other components remain the same as for the complete model.

Given the tree in Fig. 5, the flux vector of the tree permeances ϕ_{tg} reads as

$$\phi_{tg} = [\phi_{s1}, \phi_{s2}, \phi_{s3}, \phi_b, \phi_{m1}, \phi_{m2}, \phi_{a11}]^T \quad (50)$$

and the vector of the cotree fluxes is given by

$$\phi_c = [\phi_r, \phi_{l12}, \phi_{l23}, \phi_{l31}, \phi_{a12}, \phi_{a21}, \phi_{a22}, \phi_{a31}, \phi_{a32}]^T. \quad (51)$$

The magnetomotive forces are defined accordingly and the remaining fluxes and magnetomotive forces are equal to the complete model. The permeance matrices of the tree and cotree reduce to

$$\mathbf{G}_t = \text{diag}[G_{s1}, G_{s2}, G_{s3}, G_b, G_{m1}, G_{m2}, G_{a11}] \quad (52a)$$

$$\mathbf{G}_c = \text{diag}[G_r, G_{l12}, G_{l23}, G_{l31}, G_{a12}, G_{a21}, G_{a22}, G_{a31}, G_{a32}] \quad (52b)$$

and the components of the incidence matrix read as

$$\mathbf{D}_c = \begin{bmatrix} 0 & 1 & 0 & -1 & 0 & -1 & -1 & -1 & -1 \\ 0 & -1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \quad (53a)$$

$$\mathbf{D}_m = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (53b)$$

$$\mathbf{D}_g = \begin{bmatrix} 0 & -1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & -1 & -1 \\ -1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & -1 & -1 & -1 \end{bmatrix}. \quad (53c)$$

The balance and the torque equations are defined equally to the complete model and are, therefore, not repeated here. In the subsequent section, however, the derivation of the voltage equations according to Section II-D is carried out for the reduced model.

B. Voltage Equation

Following (22) the vector of unknowns $\mathbf{x}$ and the right-hand side $\mathbf{M}_1$ for the reduced permeance network of Fig. 5 are given by

$$\mathbf{x} = [u_{cs1}, u_{cs2}, u_{cs3}, \phi_{ms1}, \phi_{ms2}, u_{s1}, u_{s2}, u_{s3}, u_b, u_{m1}, u_{m2}, u_{a11}]^T \quad (54a)$$

$$\mathbf{M}_1 = [\phi_{cs1}, \phi_{cs2}, \phi_{cs3}, 0, 0, 0, 0, 0, 0, 0, 0, 0]^T. \quad (54b)$$

The column space $\mathbf{D}_c^I$ of $\mathbf{D}_c$ from (53a) reads as

$$\mathbf{D}_c^I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{bmatrix} \quad (55)$$

with the orthogonal complement $\mathbf{D}_c^\perp = [1, 1, 1]^T$. Thus, the transformation matrix $\mathbf{T}_{1c}$ according to (24) is given by

$$\mathbf{T}_{1c} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix} \quad (56)$$

and the matrix $\mathbf{H}_{1r}$, see (37) reads as

$$\mathbf{H}_{1r} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (57)$$

The linear combination of coil currents which can be calculated from the set of equations are defined by

$$\mathbf{H}_{1r} \mathbf{T}_{1c} \mathbf{u}_{tc} = \begin{bmatrix} u_{cs1} - u_{cs3} \\ u_{cs2} - u_{cs3} \end{bmatrix} \quad (58)$$

and the sum of the currents $(\mathbf{D}_c^\perp)^T \mathbf{u}_{tc} = u_{cs1} + u_{cs2} + u_{cs3}$, see (36), cannot be deduced from the permeance network. This is immediately clear, since applying the same current to all three coils does not change the fluxes in the machine.

The vector of independent coil fluxes is then given by

$$\mathbf{H}_{1r} \mathbf{T}_{1c} \boldsymbol{\phi}_{tc} = \begin{bmatrix} \phi_{cs1} - \phi_{cs3} \\ \phi_{cs2} - \phi_{cs3} \end{bmatrix} \quad (59)$$

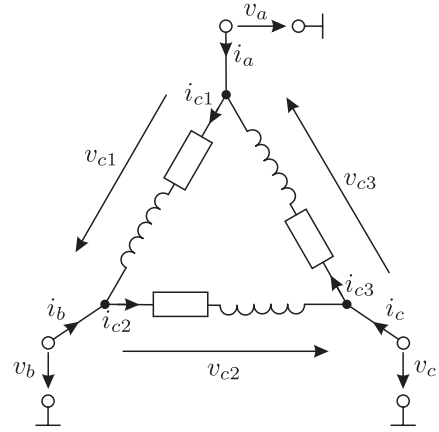


Fig. 6. Electrical connection of the motor coils (delta).

and the constraint $(\mathbf{D}_c^\perp)^T \boldsymbol{\phi}_{tc} = \phi_{cs1} + \phi_{cs2} + \phi_{cs3} = 0$ has to be met.

The coils of the motor are connected in delta connection, Fig. 6, which does not directly imply an additional constraint on the currents. The electric voltages, however, have to meet $v_{c1} + v_{c2} + v_{c3} = 0$. Using this constraint in the ode

$$N_c \frac{d}{dt} (\phi_{cs1} + \phi_{cs2} + \phi_{cs3}) = -R_c (i_{c1} + i_{c2} + i_{c3}) + v_{c1} + v_{c2} + v_{c3} \quad (60)$$

$i_{c1} + i_{c2} + i_{c3} = 0$ can be directly deduced. Finally, the set of independent differential equations is given by

$$N_c \frac{d}{dt} (\phi_{cs1} - \phi_{cs3}) = -R_c (i_{c1} - i_{c3}) + v_{c1} - v_{c3} \quad (61a)$$

$$N_c \frac{d}{dt} (\phi_{cs2} - \phi_{cs3}) = -R_c (i_{c2} - i_{c3}) + v_{c2} - v_{c3}. \quad (61b)$$

Remark 2: Note that the electrical interconnection of the coils does not have to be considered since $\mathbf{H}_{1r} \mathbf{T}_{1c} \mathbf{V}_c = \mathbf{I}$ with

$$\mathbf{V}_c = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \\ -1 & -1 \end{bmatrix} \quad (62)$$

and $\bar{\mathbf{u}}_{tc} = N_c [i_{c1} - i_{c3}, i_{c2} - i_{c3}]^T$.

C. Comparison With Complete Model and Measurements

To prove that the reduced model captures the essential behavior of the complete model with sufficient accuracy, a comparison of the torques for zero current [Fig. 7(a)] and for $-i_{c2} = i_{c3} = 2.5$ A, $i_{c1} = 0$ A [Fig. 7(b)] is given. It can be seen that almost perfect agreement between the two models can be achieved. The comparison of the magnetomotive force u_{s3} in Fig. 7(c) shows some minor differences between the complete and reduced model, which, however, do not significantly influence the torque. Therefore, the simplifications of the reduced model can be considered feasible.

For the evaluation of the model quality in comparison with the behavior of the real motor, measurements at a test bench were performed. The test bench given in Fig. 8 is composed of: 1) the PSM; 2) a torque measurement shaft; 3) a highly accurate resolver; 4) an inertia disk; and 5) a harmonic drive. The PSM is connected to a voltage source, where the terminal voltage v_c is adjusted to obtain a desired current i_c while

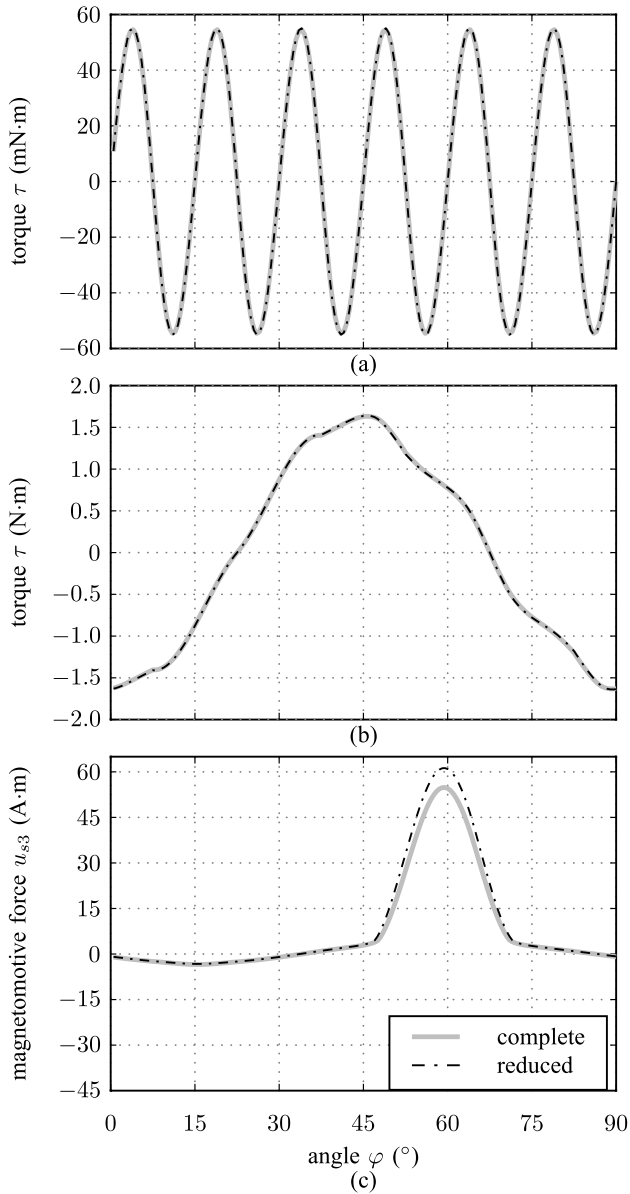


Fig. 7. Comparison of the complete with the reduced model (a) for zero currents and (b) and (c) for $-i_{c2} = i_{c3} = 2.5$ A and $i_{c1} = 0$ A.

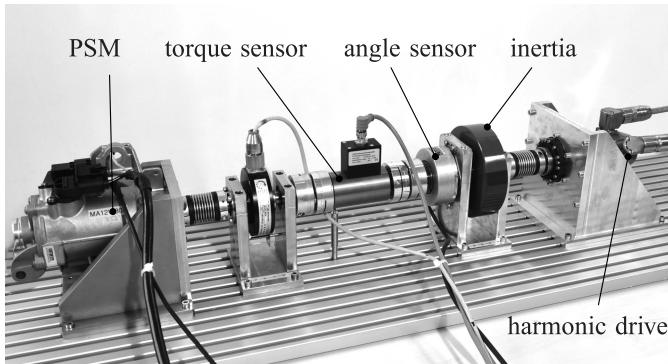


Fig. 8. Test bench for the PSM.

the terminal voltages v_a and v_b are set to zero, Fig. 6. To measure the torque τ as a function of the angle φ , the PSM is driven by a harmonic drive motor at a constant rotational speed of $n = 2$ rpm.

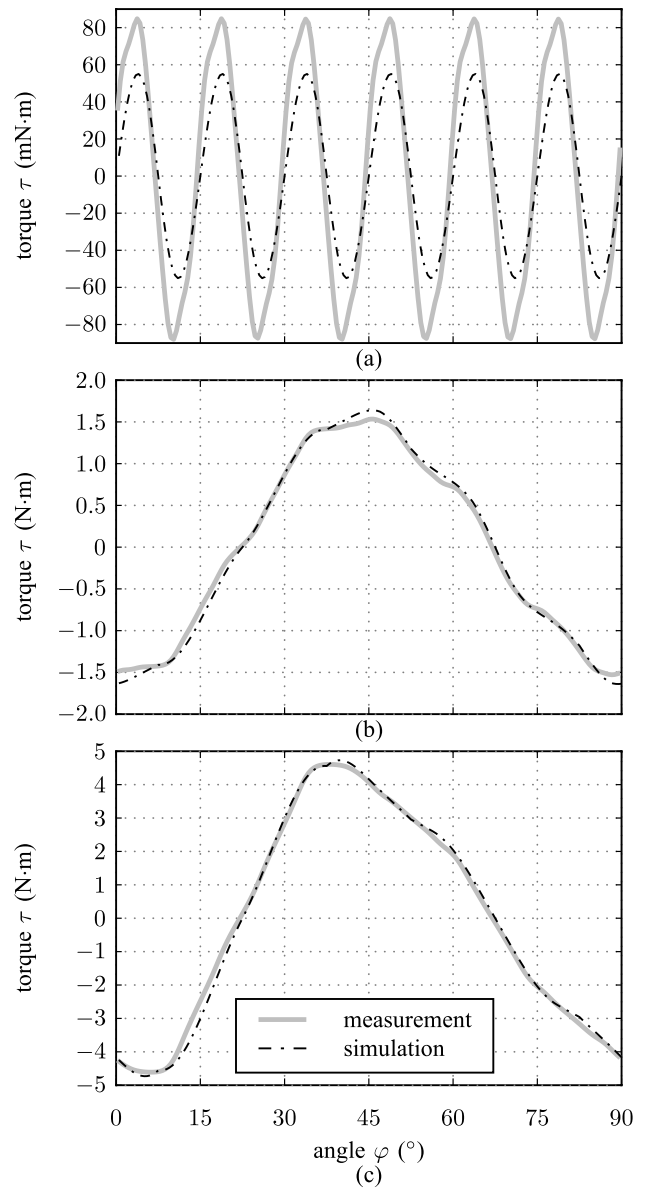


Fig. 9. Comparison of the reduced model with measurements (a) for zero currents, (b) for $-i_{c2} = i_{c3} = 2.5$ A, $i_{c1} = 0$ A, and (c) for $-i_{c2} = i_{c3} = 7.5$ A, $i_{c1} = 0$ A.

Fig. 9 depicts a comparison of the measured and simulated torque of the PSM. It can be seen that a rather good agreement between measurement and reduced model is given, which is remarkable, since the model has only been parameterized by means of geometrical and nominal material parameters.

The reduced model, however, is not accurate enough for a high precision control strategy. Therefore, the next section is concerned with the calibration of certain model parameters to further improve the model accuracy.

IV. MODEL CALIBRATION

The main reason for the model errors are the inaccuracies in the air gap permeances G_{ajk} . Thus, the following strategy is introduced for the identification of the air gap permeances.

1) The torque is measured for fixed currents $i_{cs1} = 0$, $-i_{cs2} = i_{cs3} = i_{cs}$ and a fixed step size in the angle $\Delta\varphi$, resulting in a measurement vector τ_m^k , $k = 1, \dots, N_\varphi$ with the

corresponding angles $\varphi^k = k \Delta\varphi$, the number of measurements N_φ and $\Delta\varphi N_\varphi = \pi/2$.

2) It is assumed that $G_a = G_{a,\text{nom}} + \Delta G_a$, with the nominal value $G_{a,\text{nom}}$ and the corrective term ΔG_a to be identified. Of course, the corrective term has to meet the symmetry condition (8), $\Delta G_{alm} = \Delta G_a(\varphi - (l-1)\pi/6 - (m-1)\pi/4)$, $l = 1, 2, 3$ and $m = 1, 2$. For fixed angles φ^k the corresponding values are given by $\Delta G_{alm}^{\gamma_{lm}}$, where the index γ_{lm} is defined as

$$\gamma_{lm} = \text{mod}\left(k - (l-1)\frac{N_\varphi}{3} - (m-1)\frac{N_\varphi}{2} - 1, N_\varphi\right) + 1. \quad (63)$$

3) For each angle φ^k , the relation

$$(\mathbf{G}_t + \mathbf{D}_g \mathbf{G}_c \mathbf{D}_g^T) \mathbf{u}_{tg}^k = -\mathbf{D}_g \mathbf{G}_c (\mathbf{D}_c^T \mathbf{u}_{tc} + \mathbf{D}_m^T \mathbf{u}_{tm}) \quad (64)$$

with $\mathbf{G}_t(\varphi^k, \mathbf{u}_{tg}^k)$ and $\mathbf{G}_c(\varphi^k, \mathbf{u}_{tg}^k)$ has to be fulfilled, see (18).

4) The derivation $\partial \Delta G_a / \partial \varphi$, needed for the calculation of the torque, see (20), is approximated by

$$\left. \frac{\partial \Delta G_a}{\partial \varphi} \right|_{\varphi^k + \frac{\Delta\varphi}{2}} = \frac{\Delta G_a^{k+1} - \Delta G_a^k}{\Delta\varphi}. \quad (65)$$

The corresponding magnetomotive forces of the air gap also have to be evaluated at $\varphi^k + \Delta\varphi/2$. Since they are calculated from (64) at the angles φ^k , these values are obtained by averaging the magnetomotive forces at the angles φ^k and φ^{k+1} , that is

$$u_{alm}|_{\varphi^k + \frac{\Delta\varphi}{2}} = \frac{u_{alm}^{k+1} + u_{alm}^k}{2} \quad (66)$$

with $l = 1, 2, 3$ and $m = 1, 2$.

With these prerequisites, N_φ torque equations in the form $\tau^k = \tau_m^k$ and $7N_\varphi$ nonlinear equations defined by (64), are given. The N_φ unknown values of ΔG_a^k and the $7N_\varphi$ unknown vectors $\mathbf{u}_{tg}^k$ are given as the solution of this set of equations. This solution is found numerically using, e.g., MATLAB and results in the desired corrective term ΔG_a^k as a function of φ .

Fig. 10 shows a comparison of the nominal air gap permeance $G_a(\varphi)$ model adopted from [2] with the identified values, where measurements with fixed currents $i_{cs1} = 0$, $-i_{cs2} = i_{cs3} = i_{cs} = 5$ A were used for the identification. It can be seen that the basic shape is equal to the nominal characteristics, only the maximum value is reduced and the transition phase is slightly changed. The identified shape seems to be reasonable since the changes might account for unmodeled leakages.

In Fig. 11, the torque of the calibrated model is compared with measurement results. These results show a significant improvement to the uncalibrated model in Fig. 9 and a very good agreement in the complete operating range of the motor. Thus, it can be deduced that both the inhomogeneous air gap as well as saturation in the motor are adequately represented by the proposed model. It is worth noting that an even better agreement between measurement and model could be achieved for the cogging torque if the calibration would have been performed at a lower current, e.g., $i_{cs} = 2.5$ A. Then, however, the results for high currents would be worse such that the presented results are a good compromise between the accuracy for low and high currents.

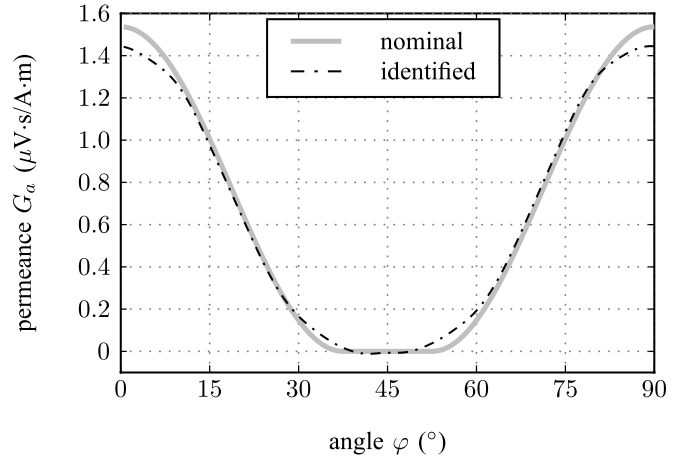


Fig. 10. Comparison of the identified and the nominal air gap permeance $G_a(\varphi)$.

The comparison of the induced voltages v_{csj} , $j = 1, 2, 3$ for a fixed angular velocity of 120 rad/s given in Fig. 12 further confirm the high accuracy of the proposed model.

In conclusion, it was shown that a calibrated permeance model in form of a state-space representation with minimum number of states is suitable for the accurate description of the behavior of the motor in the complete operating range. In the subsequent section, a classical dq0-model of the motor, as it is typically employed in the controller design of PSM, will be derived. To do so, first a magnetically linear model is extracted from the nonlinear reduced model. It will be shown that the simplifications associated with the magnetically linear and especially with the dq0-model result in rather large deviations from the measurement results. This also implies that a controller design based on dq0-models is not able to exploit the full performance of model based nonlinear control strategies.

V. SIMPLIFIED MODELS

A. Magnetically Linear Model

If it is assumed that the relative permeability μ_r of all permeances is constant, then a magnetically linear permeance model is obtained. Starting from (18) (of course using the incidence matrix $\mathbf{D}$ and the tree and cotree magnetomotive forces and fluxes of the reduced model of Section III), the magnetomotive forces $\mathbf{u}_{tg}$ of the tree permeances can be calculated in the form

$$\mathbf{u}_{tg} = -(\mathbf{G}_t + \mathbf{D}_g \mathbf{G}_c \mathbf{D}_g^T)^{-1} \mathbf{D}_g \mathbf{G}_c (\mathbf{D}_c^T \mathbf{u}_{tc} + \mathbf{D}_m^T \mathbf{u}_{tm}) \quad (67)$$

and the coil fluxes ϕ_{tc} read as, see (17)

$$\begin{aligned} \phi_{tc} = & -\mathbf{D}_c \left[\mathbf{G}_c - \mathbf{G}_c \mathbf{D}_g^T (\mathbf{G}_t + \mathbf{D}_g \mathbf{G}_c \mathbf{D}_g^T)^{-1} \mathbf{D}_g \mathbf{G}_c \right] \\ & \times (\mathbf{D}_c^T \mathbf{u}_{tc} + \mathbf{D}_m^T \mathbf{u}_{tm}). \end{aligned} \quad (68)$$

Thus, the coil fluxes are given in the form of a superposition of the flux due to the coil currents $\mathbf{u}_{tc} = N_c \mathbf{i}_c$ and the permanent magnets $\mathbf{u}_{tm}$. Inserting (68) into the voltage equation (21), we get for the left-hand side

$$N_c \frac{d}{dt} \phi_{tc} = N_c \underbrace{\frac{\partial \phi_{tc}}{\partial \varphi}}_{\mathbf{J}} \omega + N_c \underbrace{\frac{\partial \phi_{tc}}{\partial \mathbf{i}_c}}_{\mathbf{L}} \frac{d}{dt} \mathbf{i}_c. \quad (69)$$

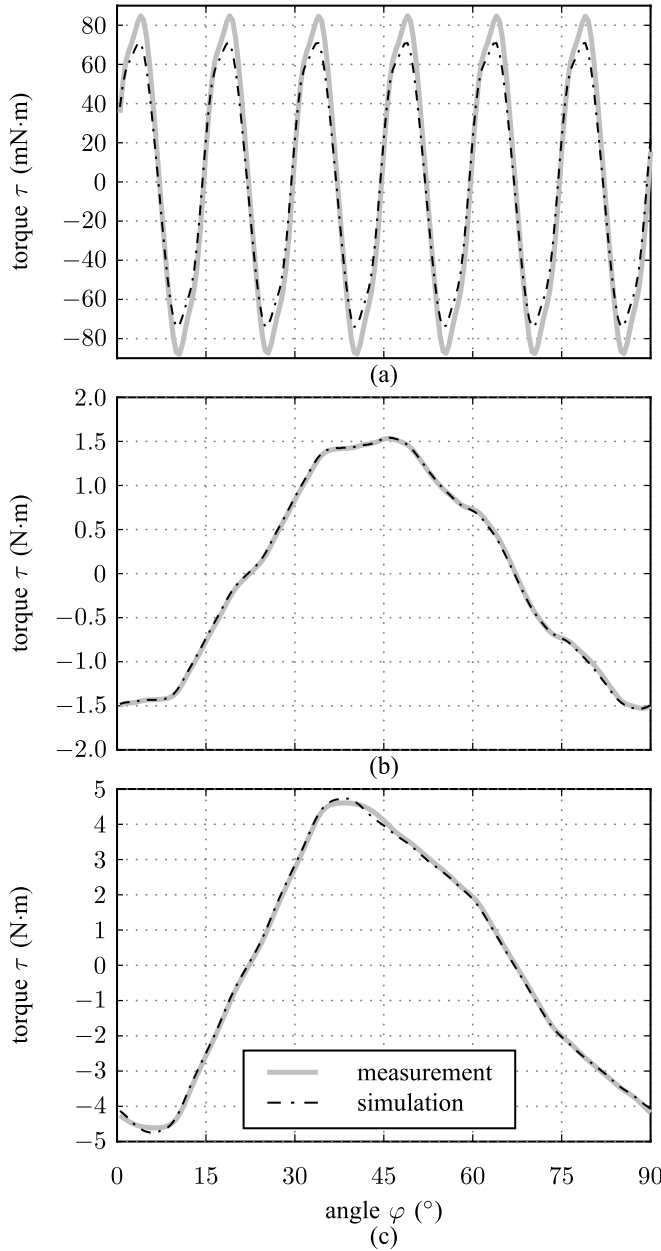


Fig. 11. Comparison of the torque of the calibrated model with measurements (a) for zero currents, (b) for $-i_{c2} = i_{c3} = 2.5$ A, $i_{c1} = 0$ A, and (c) for $-i_{c2} = i_{c3} = 7.5$ A, $i_{c1} = 0$ A.

The (symmetric) inductance matrix $\mathbf{L}$ can be formulated as

$$\mathbf{L} = -N_c^2 \mathbf{D}_c \left[\mathbf{G}_c - \mathbf{G}_c \mathbf{D}_g^T \left(\mathbf{G}_t + \mathbf{D}_g \mathbf{G}_c \mathbf{D}_g^T \right)^{-1} \mathbf{D}_g \mathbf{G}_c \right] \mathbf{D}_c^T \quad (70)$$

and the vector $\mathbf{J}$ reads as

$$\mathbf{J} = -N_c \mathbf{D}_c \mathbf{T}_J \left(\mathbf{D}_c^T \mathbf{u}_{tc} + \mathbf{D}_m^T \mathbf{u}_{tm} \right) \quad (71)$$

with

$$\mathbf{T}_J = \left[\frac{\partial \mathbf{G}_c}{\partial \varphi} - \frac{\partial \mathbf{G}_c}{\partial \varphi} \mathbf{D}_g^T \mathbf{H}_5 \mathbf{D}_g \mathbf{G}_c - \mathbf{G}_c \mathbf{D}_g^T \mathbf{H}_5 \mathbf{D}_g \frac{\partial \mathbf{G}_c}{\partial \varphi} + \mathbf{G}_c \mathbf{D}_g^T \mathbf{H}_5 \left(\frac{\partial \mathbf{G}_t}{\partial \varphi} + \mathbf{D}_g \frac{\partial \mathbf{G}_c}{\partial \varphi} \mathbf{D}_g^T \right) \mathbf{H}_5 \mathbf{D}_g \mathbf{G}_c \right] \quad (72)$$

and

$$\mathbf{H}_5 = \left(\mathbf{G}_t + \mathbf{D}_g \mathbf{G}_c \mathbf{D}_g^T \right)^{-1}. \quad (73)$$

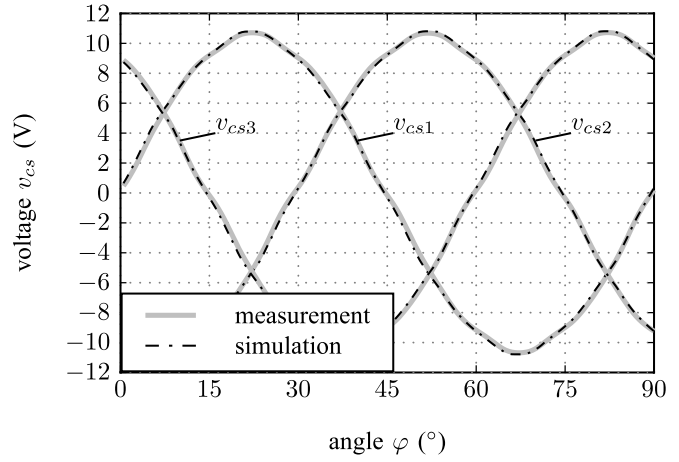


Fig. 12. Comparison of the induced voltages of the calibrated model with measurements for $\omega = 120$ rad/s.

Given these results, the voltage equation (21) can be formulated in the well-known form

$$\mathbf{L}(\varphi) \frac{d}{dt} \mathbf{i}_c = -\mathbf{J}(\varphi) \omega + \mathbf{R}_c \mathbf{i}_c - \mathbf{v}_c. \quad (74)$$

Note that the inductance matrix $\mathbf{L}$ and the vector $\mathbf{J}$ are both nonlinear functions of the rotor angle φ . According to (20), the torque τ of the motor in the magnetic linear case is given by

$$\begin{aligned} \tau = & \underbrace{\frac{1}{2} p \mathbf{u}_{tc}^T \mathbf{D}_c \mathbf{T}_J \mathbf{D}_c^T \mathbf{u}_{tc}}_{\tau_r} + \underbrace{\frac{1}{2} p \mathbf{u}_{tm}^T \mathbf{D}_m \mathbf{T}_J \mathbf{D}_m^T \mathbf{u}_{tm}}_{\tau_c} \\ & + \underbrace{p \mathbf{u}_{tm}^T \mathbf{D}_m \mathbf{T}_J \mathbf{D}_c^T \mathbf{u}_{tc}}_{\tau_p}. \end{aligned} \quad (75)$$

Here, three different parts can be distinguished: 1) for zero coil currents, i.e., $\mathbf{u}_{tc} = \mathbf{0}$, the remaining part τ_c represents the cogging torque of the motor. 2) Excluding the permanent magnets of the motor, i.e., setting $\mathbf{u}_{tm} = \mathbf{0}$, only the reluctance torque τ_r due to the inhomogeneous air gap is present. 3) The part τ_p represents the main part of the torque. It is the only part which can be found in an ideal PSM with a homogenous air gap.

B. Fundamental Wave Model

The magnetically linear model of the previous section still covers the complete nonlinearity due to the air gap permeances. In this subsection, a further simplification is made, where only the average values and the fundamental wave components of the corresponding parts are considered.

Applying this approach to (70), the inductance matrix is given by

$$\mathbf{L} = \begin{bmatrix} L_m & -\frac{1}{2} L_m & -\frac{1}{2} L_m \\ -\frac{1}{2} L_m & L_m & -\frac{1}{2} L_m \\ -\frac{1}{2} L_m & -\frac{1}{2} L_m & L_m \end{bmatrix} \quad (76)$$

with the constant main inductance L_m . The term $\mathbf{J}$ reduces to

$$\mathbf{J}(\varphi) = \hat{J} \begin{bmatrix} \sin(p\varphi) \\ \sin\left(p\varphi - \frac{2}{3}\pi\right) \\ \sin\left(p\varphi - \frac{4}{3}\pi\right) \end{bmatrix} \quad (77)$$

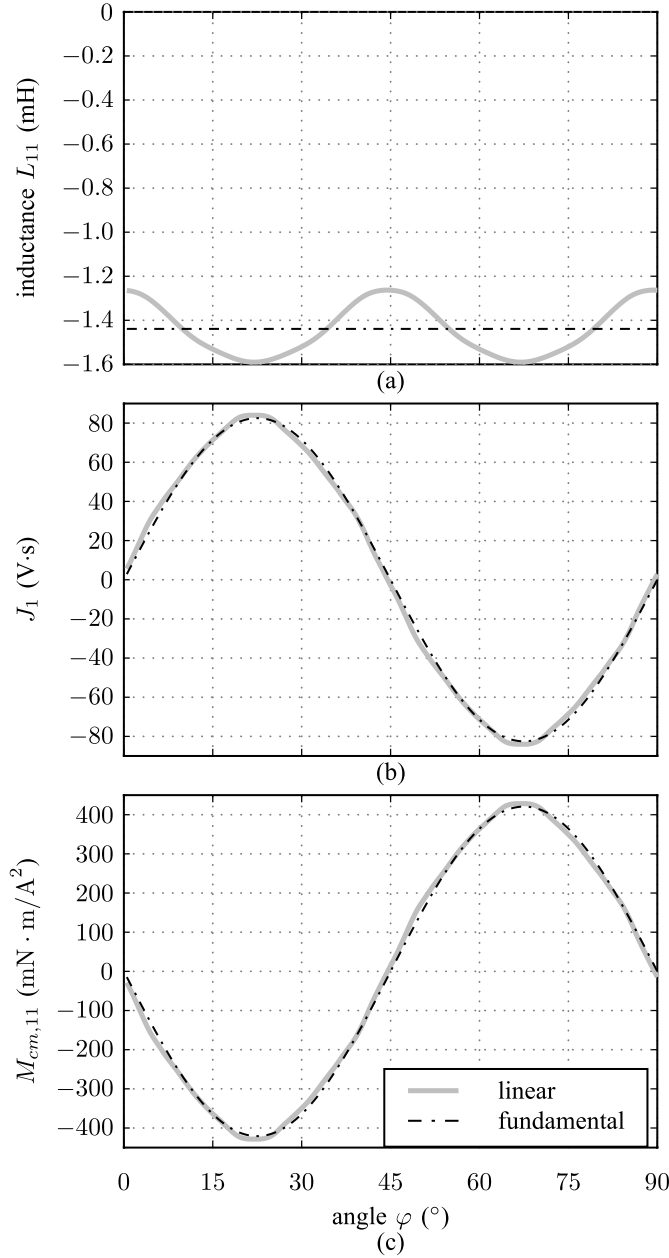


Fig. 13. (a) Entry L_{11} of the inductance matrix $\mathbf{L}$, (b) J_1 of the vector $\mathbf{J}$, and (c) $M_{cm,11}$ of the matrix $\mathbf{M}_{cm}$ for the magnetically linear and the fundamental wave model.

and the torque can be formulated as $\tau = p \mathbf{u}_{tm}^T \mathbf{M}_{cm}(\varphi) \mathbf{u}_{tc}$, where $\mathbf{M}_{cm}(\varphi)$ reads as

$$\hat{M} \begin{bmatrix} \sin(p\varphi) & \sin(p\varphi - \frac{2\pi}{3}) & \sin(p\varphi - \frac{4\pi}{3}) \\ -\sin(p\varphi) & -\sin(p\varphi - \frac{2\pi}{3}) & -\sin(p\varphi - \frac{4\pi}{3}) \end{bmatrix}. \quad (78)$$

The coefficients L_m , $\hat{J}$, and $\hat{M}$ can be obtained, e.g., by a Fourier analysis of the corresponding entries of the magnetically linear model. Fig. 13 shows a comparison of the entries of the inductance matrix $\mathbf{L}$, the vector $\mathbf{J}$, and the matrix $\mathbf{M}_{cm}$ between the magnetically linear model and the fundamental wave model. It can be seen that a rather good approximation of the magnetically linear model can be obtained by means of the fundamental wave model.

The well-known dq0-representation of the fundamental wave model can be found using the transformations

$$\begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} = \mathbf{K}(\varphi) \begin{bmatrix} i_{c1} \\ i_{c2} \\ i_{c3} \end{bmatrix}, \quad \begin{bmatrix} v_d \\ v_q \\ v_0 \end{bmatrix} = \mathbf{K}(\varphi) \begin{bmatrix} v_{c1} \\ v_{c2} \\ v_{c3} \end{bmatrix} \quad (79)$$

with the transformation matrix $\mathbf{K}(\varphi)$

$$\mathbf{K}(\varphi) = \begin{bmatrix} \cos(p\varphi) & \cos(p\varphi - \frac{2\pi}{3}) & \cos(p\varphi - \frac{4\pi}{3}) \\ \sin(p\varphi) & \sin(p\varphi - \frac{2\pi}{3}) & \sin(p\varphi - \frac{4\pi}{3}) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}. \quad (80)$$

Then, the dq0-model takes the form

$$\frac{d}{dt} i_d = \frac{2}{3} \frac{1}{L_m} \left(-\frac{3}{2} L_m p \omega i_q + R_c i_d - v_d \right) \quad (81a)$$

$$\frac{d}{dt} i_q = \frac{2}{3} \frac{1}{L_m} \left(\frac{3}{2} L_m p \omega i_d - \frac{3}{2} \hat{J} \omega + R_c i_q - v_q \right) \quad (81b)$$

and the torque is given by

$$\tau = 2p \hat{M} u_{ms1} N_c i_q. \quad (82)$$

C. Comparison of the Models

Up to now three models of different complexity, i.e., a magnetically nonlinear model, a magnetically linear and a fundamental wave model, were presented in this paper. In this section, the torque τ calculated by these models is compared with measurement results, see Fig. 14.

The results for zero current [Fig. 14(a)] show that the cogging torque can be reproduced rather well by the nonlinear model. Even the magnetically linear model shows the basic behavior of the cogging torque, however, with larger errors compared to the nonlinear model. As a matter of fact, it is not possible to reproduce the cogging torque with the fundamental wave model. Thus, this model gives the worst results as it was, of course, expected.

For nominal and high currents shown in Fig. 14(b) and (c), respectively, this result is confirmed. Again the nonlinear model gives excellent agreement with the measurements while the performance of the magnetically linear model degrades with increasing currents. This results from the fact that saturation is not included in the magnetically linear model. The basic shape is, however, much better reproduced than in the fundamental wave model.

This brief comparison shows that a controller designed using a fundamental wave model cannot systematically account for the cogging torque and saturation. Using instead the nonlinear model for a controller design it can be expected that the control performance is superior to controllers based on fundamental wave models. The obvious drawback of the nonlinear model is the increased complexity of the resulting control strategy. Here, the magnetically linear model might be a good compromise between model complexity and model accuracy for the controller design and will yield significant improvements in comparison to fundamental wave models.

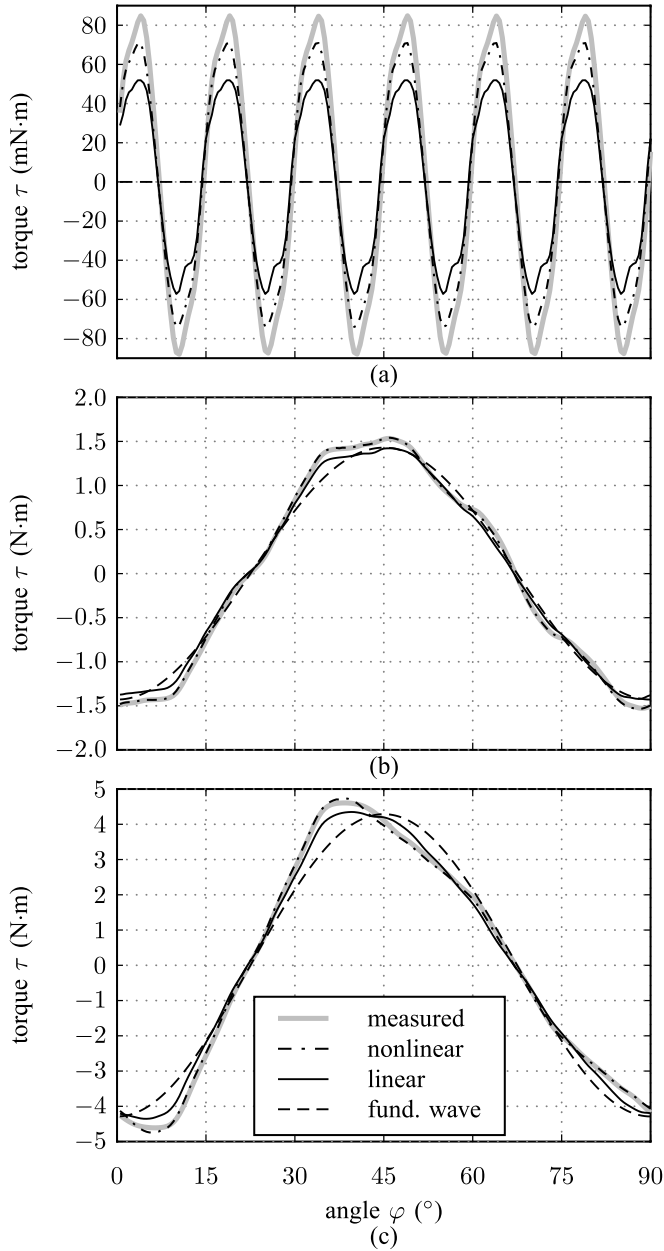


Fig. 14. Comparison of the measurement results with the nonlinear, the magnetically linear and the fundamental wave model for (a) for zero currents, (b) $-i_{c2} = i_{c3} = 2.5$ A, $i_{c1} = 0$ A, and (c) for $-i_{c2} = i_{c3} = 7.5$ A, $i_{c1} = 0$ A.

VI. CONCLUSION

A systematic modeling framework for PSM with internal magnets was outlined in this paper. Different from existing works, the balance equations were derived based on graph-theory, which allows for a systematic calculation of the minimum number of nonlinear equations. Furthermore, the choice of a suitable state and the systematic consideration of the electrical connection of the coils were discussed. The quality of the calibrated model was shown by a comparison with measurement results. Finally, a magnetically linear and a dq0-model have been derived and compared with the nonlinear model.

Future work will deal with the application of the methodology to other motor designs as, e.g., PSM with surface magnets, reluctance machines or asynchronous machines. In addition,

the use of the models derived in this paper for nonlinear and optimal controller design is an ongoing topic of research. Here, first results show a high potential of the modeling approach and a significant improvement in comparison to control strategies using classical dq0-models.

APPENDIX

EXISTENCE AND UNIQUENESS OF SOLUTION

The set of nonlinear equations in (18) has to be solved numerically. Thus, it is interesting to examine if a solution exists and if it is unique. The matrices $\mathbf{G}_r$ and $\mathbf{G}_c$ are positive semidefinite matrices for all $\mathbf{u}_{tg}$ and φ . This can be easily seen since the entries of these diagonal matrices are positive except for the air gap permeances, which can become zero for certain angles φ . In addition, a suitable construction of the permeance network ensures that $\mathbf{D}_g$ has independent rows such that $\mathbf{D}_g \mathbf{G}_c \mathbf{D}_g^T$ is also positive semidefinite. To show that the sum of this term with $\mathbf{G}_r$ is even positive definite, consider the vector $\mathbf{x}$ which fulfills

$$\mathbf{x}^T \mathbf{G}_r \mathbf{x} = 0. \quad (83)$$

The only possible solution $\mathbf{x}$ of (83) is equal to $\mathbf{x} = [0, \dots, \alpha]^T$, $\alpha \in \mathbb{R}$. It is then rather simple to show that

$$\mathbf{x}^T \mathbf{D}_g \mathbf{G}_c \mathbf{D}_g^T \mathbf{x} > 0 \quad \forall \mathbf{u}_{tg}, \varphi \quad (84)$$

which implies that $\mathbf{F} = \mathbf{G}_r + \mathbf{D}_g \mathbf{G}_c \mathbf{D}_g^T$ is positive definite. In the magnetic linear case, the permeances are independent of the magnetomotive force and therefore, the positive definiteness of $\mathbf{F}$ is sufficient for the existence and uniqueness of a solution of (18). In the nonlinear case, however, it has to be shown the Jacobian of $\mathbf{F}(\mathbf{u}_{tg}, \varphi) \mathbf{u}_{tg}$ is positive definite [39]. The Jacobian can be written in the form

$$\mathbf{F} + \sum_{j=1}^n \frac{\partial \mathbf{F}}{\partial u_{tg,j}} \mathbf{u}_{tg} \quad (85)$$

where $u_{tg,j}$ describes the j th entry of $\mathbf{u}_{tg}$. Using the fact that

$$\mu_r(H) + H \frac{\partial \mu_r(H)}{\partial H} > 0 \quad (86)$$

holds, it can be shown that the Jacobian (85) indeed is positive definite for all $\mathbf{u}_{tg}$ and φ . It can be further shown that

$$\lim_{\|\mathbf{u}_{tg}\| \rightarrow \infty} \|\mathbf{F}(\mathbf{u}_{tg}) \mathbf{u}_{tg}\| = \infty \quad (87)$$

holds, which implies that there exists a unique solution of the set of nonlinear (18), see [39].

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